Program 12: Build and Deploy a Complete DevOps Pipeline, Discussion on Best Practices and Q&A.

- Build Pipeline (CI) compiles the code, runs tests, and prepares an artifact (e.g., .jar, .zip) when code is pushed.
- Release Pipeline (CD) takes that artifact and deploys it to a target environment (e.g., Azure App Service).
- Together, it ensures smooth, continuous delivery from code commit to deployment with minimal manual steps.

Build Pipeline (CI) Setup:

Step 1: Go to Azure DevOps → Pipelines → Create Pipeline

Step 2: Select GitHub as the code repository and choose your sample Maven project.

Step 3: Choose **YAML** and create the following build steps:

- Checkout code
- Maven build task (mvn clean install)
- Publish artifact (e.g., JAR or ZIP)

Step 4: Commit and run the pipeline.

Step 5: If **Hosted Parallelism is not approved**, you may simulate steps using:

```
script: echo "Simulating Maven Build...
script: echo "Simulating Artifact Publish...
```

Release Pipeline (CD) Setup:

Step 6: Go to Pipelines → **Releases** → **New Pipeline**

Step 7: Add an Artifact → Select the build pipeline output.

Step 8: Add a Stage → Choose "**Empty Job**"

Step 9: In the Tasks tab, add a Command Line Script task:

```
echo "Simulating deployment to Azure App Service...
echo "Deploying artifact: HelloWorld-1.0.jar
```

Step 10: Save and create a release → Manually trigger it to run.

Step 11: Use screenshots to validate artifact linkage and simulation.

Simulating Secrets with Variable Groups:

Step 12: Go to Pipelines → Library → + Variable Group

Step 13: Name it (e.g., MySecretsGroup), add secure variables (e.g., DB_PASSWORD)

Step 14: Save the group and link it in your release pipeline (under Variables tab).

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Discussion on Best Practices (Examples)

- Use version control for all YAML/pipeline files.
- Simulate builds when agents are unavailable.
- Store secrets securely in variable groups or Key Vault.
- Break pipelines into clear stages (build, test, deploy).
- Always review logs and test locally before release.